

# RUIWEN WANG

PH.D. CANDIDATE | AI SYSTEMS | HPC | ML SYSTEMS

Sorbonne University & EURECOM | CIFRE with Huawei Paris Research Center |

Expected graduation: end of 2026

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## PROFILE

Ph.D. (CIFRE) candidate in Computer Science at Sorbonne University and EURECOM, in collaboration with Huawei Paris Research Center (expected end of 2026). I design symbolic, optimisation-based planners that turn large-scale hybrid-parallel LLM/DNN training into a predictable, retargetable decision problem — synthesising memory-feasible, communication-aware strategies and pipeline schedules validated on production Ascend clusters up to 16,384 NPUs. Output so far: four first-author papers (CLUSTER, Euro-Par Best Poster, NPC, HPC Asia), one under submission, co-authored large-scale MoE training, and three patent families, several transferred to production.

## DOCTORAL RESEARCH

**Thesis** — *Systematic and Portable Optimisation of Hybrid Parallelism for Large-Scale Distributed Training* (Sorbonne University, doctoral school EDITE ED 130). A portable, behaviour-level planning workflow built on a single per-layer (Transformer) cost unit that spans all seven parallelism dimensions — data, tensor, pipeline, virtual-pipeline, sequence, expert, and optimiser — and makes large-scale hybrid-parallel planning **predictable** (Prism), **interoperable** (BMPipe + H2O), **evolvable** (Concord), and **adaptable** (ManuMatic). The unit separates model, hardware, and parallelisation descriptions so a plan can be predicted before execution and retargeted across GPU/NPU clusters. Validated across model families — Mixtral-8×7B, Llama3-8B, Qwen2.5-72B, DeepSeek-141B — on Ascend NPU clusters from tens to 16,384 devices, including long-sequence regimes.

## SELECTED SYSTEMS & RESULTS

- **Prism** — **predictability via symbolic memory modeling** (HPC Asia 2026). A profiling-free, layer-level cost surface that predicts per-stage **peak memory** (and communication) analytically from the model, training configuration, and hardware description, turning memory feasibility from trial-and-error into a closed-form, retargetable check across model families and clusters.
- **BMPipe** — **pipeline efficiency** (IEEE CLUSTER 2025). A symbolic bubble model (recomputation / stage-imbalance / preparation bubbles) coupled with an ILP that jointly sets stage boundaries, intra-stage chunking, and per-layer recomputation to drive down pipeline idle time; up to  $1.36\times$  throughput at 83–92% HBM utilization, validated to 16,384 Ascend NPUs.
- **H2O** — **holistic, all-dimension joint optimisation** (Euro-Par 2025, Best Poster Award). Co-optimises parallelism degrees across all dimensions together with micro-batch size and adaptive recomputation in one workflow, instead of tuning each knob separately; raises MFU 26.13%→35.73% on DeepSeek-141B and transferred to production training.
- **Concord** — **search efficiency through reuse** (under submission, 2026). Dependency-keyed per-layer signatures pinpoint exactly which planning work a model or configuration edit invalidates, so re-planning reuses the still-valid majority instead of restarting, cutting redundant search during iterative model–strategy co-design.
- **ManuMatic** — **adaptability to new operators and runtime effects** (NPC 2025). Injects user/runtime-pinned operator shardings as first-class constraints into automatic search, keeping plans robust when new operators or effects unseen by the abstraction appear, and recovering strong plans where pure auto-search degrades (e.g. +30.1% over an expert-tuned plan on Qwen2.5-72B).

## EXPERIENCE

### Distributed and Parallel Computing Lab, Huawei Paris Research Center

Paris, France

*Ph.D. Candidate (CIFRE) and Research Engineer – LLM Training Systems*

2021 – end 2026

- Built cost-model-driven planners for hybrid and pipeline-parallel LLM training: stage partitioning, pipeline scheduling, and recomputation selection under joint memory and performance constraints.
- Designed symbolic peak-memory estimators and collective-cost models, with overlap-aware strategies for throughput and MFU tuning at large scale.
- Translated prototypes into deployable planning workflows with engineering and customer teams: configuration guidance, debuggable traces, and regression benchmarks across model families and scales.
- Drove research-to-production transfer onto production-scale training, including deployment feedback on customer workloads and long-sequence training (recognised by China Telecom for DeepSeek-MoE optimisation).
- Modelled MoE / expert-parallel and long-sequence training: expert-ownership constraints, load balancing, and sequence / virtual-pipeline scheduling.

*Role progression: Research Apprentice (2021–2022) -> Research Engineer (2022–2023) -> CIFRE Ph.D. Candidate (2023–end 2026).*

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| PRODUCTION<br>IMPACT &<br>COLLABORATION | <ul style="list-style-type: none"> <li>Transferred planner-derived strategies to production training: about <math>1.1\times</math> end-to-end on 4,096-device runs and up to <math>2.37\times</math> on long-sequence training at 512 devices.</li> <li>Contributed to large-scale DeepSeek-MoE training optimisation in deployment, with feedback loops on real customer workloads and ultra-scale clusters.</li> <li>Worked across research, engineering, and customer-facing platform teams; built reproducible benchmarking and debuggable traces that kept planner decisions trustworthy in operation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |
| PUBLICATIONS                            | <ul style="list-style-type: none"> <li><b>Ruiwen Wang</b>, Chong Li, Thibaut Tachon, Raja Appuswamy, Teng Su. <i>BMPipe: Bubble-Memory Co-optimization Strategy Planner for Very-Large DNN Training</i>. IEEE CLUSTER 2025.</li> <li><b>Ruiwen Wang</b>, Chong Li, Raja Appuswamy, Yujie Yuan. <i>H2O: Holistic Hyperparameter Optimization for Large-Scale Deep Neural Network Training</i>. Euro-Par 2025. <b>Best Poster Award</b>.</li> <li><b>Ruiwen Wang</b>, Chong Li, Hongxing Wang, Raja Appuswamy, Yujie Yuan. <i>ManuMatic: Strategy Injection for Robust Automatic Hybrid Parallelism in Distributed DNN Training</i>. NPC 2025.</li> <li><b>Ruiwen Wang</b>, Philippe Fang, Chong Li, Thibaut Tachon, Raja Appuswamy. <i>PRISM: Profiling-Free Symbolic Memory-Driven Strategy Planner for Large DNN Model Training</i>. HPC Asia 2026.</li> <li><b>Ruiwen Wang</b>, Thibaut Tachon, Chong Li, Raja Appuswamy. <i>Can Hybrid-Parallel Planning Support Alternating Model-Strategy Design? Dependency-Keyed Per-Layer Primitives for Re-Planning</i>. Under submission, 2026.</li> <li>Zhengdao Yu, <b>Ruiwen Wang</b>, Nelson Lossing, Chong Li. <i>MLLM Pipeline Bubble Modeling for Large-Scale Training</i>. HPC Asia 2026.</li> <li>Xinzhang Liu, Chao Wang, Zhihua Yang, Zhuo Jiang, <b>Ruiwen Wang</b>, et al. <i>Training Report of TeleChat3-MoE</i>. arXiv:2512.24157, 2025.</li> </ul> |  |  |
| PATENTS                                 | <ul style="list-style-type: none"> <li>Chong Li, Pierre Leca, Thibaut Tachon, <b>Ruiwen Wang</b>, Haoran Wang. <i>Devices and methods for generating execution plans for a machine learning model</i>. (WO 2025/020165 A1)</li> <li>Chong Li, <b>Ruiwen Wang</b>, Haoran Wang, Fan Yu. <i>A load balancing method to improve large model performance</i>.</li> <li>Chong Li, Thibaut Tachon, <b>Ruiwen Wang</b>, Haoran Wang. <i>A multi-dimension parallelism configuration method for large language model training</i>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |
| EDUCATION                               | <p><b>Sorbonne University &amp; EURECOM</b> <span>Paris, France</span></p> <p><i>Ph.D. in Computer Science – doctoral school EDITE (ED 130)</i> <span>2023 – end 2026</span></p> <ul style="list-style-type: none"> <li>Thesis: Systematic and Portable Optimisation of Hybrid Parallelism for Large-Scale Distributed Training; supervisors Prof. Raja Appuswamy and Prof. Chong Li.</li> </ul> <p><b>Sorbonne University</b> <span>Paris, France</span></p> <p><i>M.Sc. in Computer Science – Algorithms, Software Science and Technology</i> <span>2019 – 2022</span></p> <ul style="list-style-type: none"> <li>Supervisors: Prof. Emmanuel Chailloux and Prof. Jacques Malenfant.</li> </ul> <p><b>Paris-Saclay University</b> <span>Paris, France</span></p> <p><i>B.Sc. in Computer Science</i> <span>2016 – 2019</span></p> <ul style="list-style-type: none"> <li>Early research mentored by Prof. Isabelle Guyon, Dr. Viviane Pons, and Dr. Zhengying Liu.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |
| KEYWORDS                                | Hybrid parallelism · pipeline scheduling · MoE / expert parallelism · symbolic cost & peak-memory modeling · ILP / MILP planning · recomputation · communication overlap · MFU optimisation · Ascend NPU · portability / retargeting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |
| AWARDS                                  | <ul style="list-style-type: none"> <li><b>Golden Prize</b>, Huawei Challenge Award for Automatic Load Balancing <span>2024.08</span></li> <li><b>Best Poster Award</b>, Euro-Par conference <span>2025.08</span></li> <li><b>Letter of Appreciation</b>, China Telecom for DeepSeek-MoE training optimization deployment <span>2025.12</span></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |
| SKILLS                                  | <p><b>Parallelism:</b> data, tensor, pipeline, virtual-pipeline, sequence, expert, optimiser</p> <p><b>Optimisation:</b> ILP / MILP, constraint programming, symbolic cost &amp; peak-memory modeling</p> <p><b>Programming:</b> Python, C++, Java</p> <p><b>ML stacks:</b> MindSpore, PyTorch; Ascend NPU / GPU clusters (to 16k), cloud</p> <p><b>Tooling:</b> Ascend profiling, tracing, performance regression benchmarking; Linux, Git</p> <p><b>Languages:</b> Chinese (native), English (fluent), French (fluent), Cantonese</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |